

EDA - PROJECT SUMMARY

HealthConnect Appointment Attendance and No-Show Analysis

Dataset: HealthConnect Appointment Data

By

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WHAT WAS PLANNED

The main objective for Week 5 was to conduct exploratory data analysis of the HealthConnect Appointment Dataset, develop selected KPIs, create analytical visualisations, identify meaningful business insights, and provide recommendations that could support the reduction of missed appointments.

The plan also included reviewing data quality, documenting assumptions and limitations, testing initial analytical outputs, and identifying opportunities for cross-track collaboration.

WHAT WAS COMPLETED

The planned Week 5 activities were completed.

- Data preparation and quality checks were performed, including checks for missing values, duplicates, data types, valid ranges, and categorical consistency.
- The dataset was explored across relevant patient, appointment, booking, historical, reminder, and operational variables.
- A total of 13 analytical visualisations were developed to investigate appointment outcomes and potential patterns associated with attendance, no-shows, and cancellations.
- Four KPIs were calculated:
 - ✓ No-Show Rate: 48.46%
 - ✓ Attendance Rate: 46.28%
 - ✓ Repeat No-Show Rate: 55.41%
 - ✓ Reminder Attendance Rate: 47.63%
- A two-page Power BI analytical dashboard was developed to present the KPIs and exploratory findings.

KEY FINDINGS/DEVELOPMENT OUTCOMES

The exploratory analysis provided an overall view of appointment attendance and no-show patterns within the HealthConnect Appointment Dataset. Several variables were identified as relevant areas for understanding appointment

The key development outcomes were:

1. No-shows represent the largest appointment outcome
2. Previous no-show history is a strong risk indicator
3. Reminder use does not eliminate the no-show problem
4. SMS shows the strongest attendance pattern among reminder channels
5. Longer booking lead times are associated with more no-show activity
6. Longer travel distance and waiting time is associated with lower attendance

MAJOR CHALLENGES ENCOUNTERED

One of the main challenges was ensuring that date fields were correctly recognised by Power BI. The appointment date and booking date fields initially contained text values and required conversion to the appropriate date format.

Another challenge involved ensuring that categorical values and appointment days were displayed consistently and in the correct logical order in the visualisations. The booking lead days, age group, waiting time and appointment-day ordering issue was resolved during data preparation.

Additional attention was required when working with variables such as waiting time and distance to ensure that their missing values were accounted for during the analysis.

IMPORTANT DECISIONS MADE

The exploratory analysis focused on relevant outcome-based visualisations rather than creating unnecessary additional variables. The original dataset was kept separate from the transformed analytical data to prevent the original records from being overwritten and to maintain a clear distinction between the source data and the prepared analytical dataset.

The analysis also focused on identifying patterns and areas for further investigation rather than interpreting exploratory relationships as proof of causation.

CHANGES FROM THE WEEK 4 APPROACH

The Week 4 approach focused primarily on defining the problem, reviewing the data dictionary, identifying relevant variables, establishing business questions, and proposing KPIs. In Week 5, this approach was extended into practical data preparation, exploratory analysis, KPI calculation, visualisation development, initial output testing, and dashboard creation.

The KPIs proposed during Week 4 were therefore implemented and calculated using the HealthConnect dataset, allowing the project to progress from planning and problem definition to actual analytical development.

CROSS-TRACK COLLABORATION

A collaboration point with the Data Science track was identified around using the analytical findings and relevant variables as a foundation for potential future predictive modelling of appointment no-shows.

The Week 5 exploratory analysis can help identify potentially useful variables and patterns that may be considered during subsequent data science activities. This creates an opportunity for the

Data Analytics and Data Science tracks to build on the same HealthConnect problem from complementary perspectives.

REMAINING WORK

There is no major outstanding analytical task required for the Week 5 stage. The remaining work will be carried forward into Week 6 and will build on the foundation established during this stage.

PROPOSED FOCUS FOR WEEK 6

The proposed focus for Week 6 is to build on the analytical foundation developed during Week 5. Depending on the direction of the HealthConnect project, this may include:

- Deeper investigation of the patterns identified during the exploratory analysis
- Refinement and further testing of analytical outputs, preparation of relevant variables for predictive modelling
- Continued collaboration with other project tracks.

The Week 5 outputs will therefore serve as a clear and usable foundation for the next stage of the HealthConnect Experience Lab.